

Study of Different Properties of Acetic Acid (Ethanoic Acid)**Aim**

To study the following properties of acetic (ethanoic) acid:

1. (1) Odour
2. (2) Solubility in water
3. (3) Effect on litmus
4. (4) Reaction with sodium bicarbonate

Requirements

Test tubes, a test tube stand, a cork, a delivery tube, a beaker, etc.

Chemicals Required

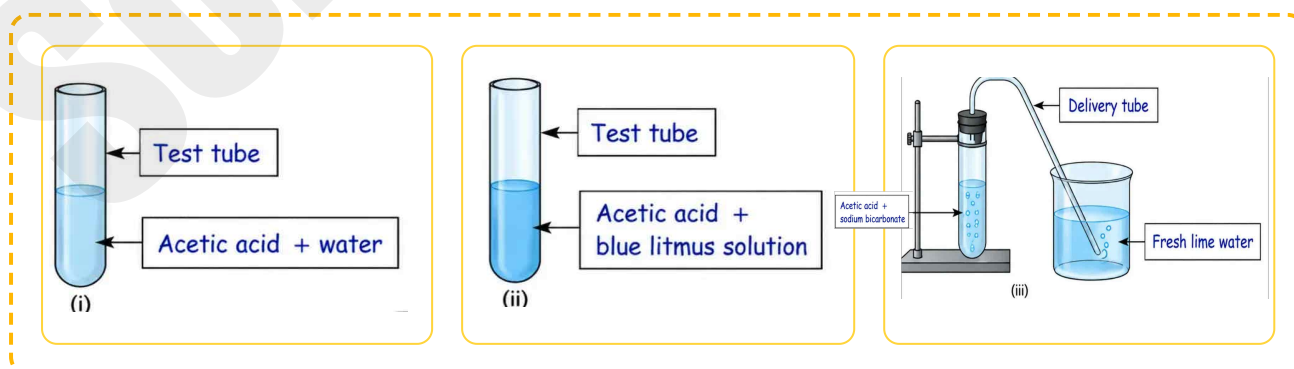
Acetic acid, red and blue litmus solution, sodium bicarbonate solution, distilled water, and fresh lime water.

Procedure

1. Take about 10 ml of acetic acid in a clean and dry test tube, and observe its colour and odour.
2. Add about 10 ml of water to the same test tube and shake it well. Observe the solution formed.
3. Add 2 to 3 drops of blue litmus solution to acetic acid taken in a separate test tube, and repeat the same with red litmus solution.
4. Take about 10 ml of acetic acid in a test tube and add a few drops of sodium bicarbonate solution to it. Pass the gas produced through fresh lime water.

Diagrams

(Label the different parts in the given diagram)



Observation and Inference Table

Sr. No.	Test	Observation	Inference
1.	Odour	Peculiar odour of vinegar	Acetic acid has odour like vinegar
2.	Solubility	Soluble in water	Acetic acid is miscible with water
3.	Blue litmus solution test Red litmus solution test	Blue colour of litmus solution changes to red No change in colour	Acetic acid is acidic in nature
4.	Reaction with sodium bicarbonate solution Pass the gas through fresh lime water	A colourless, odourless gas is liberated with brisk effervescence Lime water turns milky. After passing more gas it becomes clear.	Acetic acid liberates carbon dioxide gas on reacting with sodium bicarbonate solution.

Multiple Choice Questions

1. Ethanoic (Acetic) acid

- (A) is odourless (B) has a smell of ammonia
(C) has smell of rotten eggs (D) has a vinegar smell

Answer: (D) has a vinegar smell

2. Acetic (Ethanoic) acid

- (A) turns red litmus blue (B) has pungent odour
(C) is red in colour (D) is odourless

Answer: (B) has pungent odour

3. When sodium bicarbonate solution is added to dilute acetic acid

- (A) CO_2 gas is evolved
(B) a solid settles at the bottom
(C) the mixture becomes warm
(D) the colour of the mixture becomes yellow

Answer: (A) CO_2 gas is evolved

4. is liberated when acetic acid reacts with sodium metal.

- (A) H_2 (B) O_2 (C) CO_2 (D) NH_3

Answer: (A) H_2

5. Ethanoic acid which solidifies below 290K is known as

- (A) ice (B) solid acetic acid
(C) glacial acetic acid (D) ethyl acetate

Answer: (C) glacial acetic acid

Remember

Ethanoic acid is a colourless liquid with boiling point 118°C . Ethanoic acid is commonly known as acetic acid.

Date: _____

Teacher's Signature: _____